GET it digital

Modul 3:

Elektrische Bauelemente

Prof. Dr.-Ing. Michael Hillgärtner
Sven Micun, B.Eng.
Filimon Stergianos

A collaborative project recommended by

sponsored by

Stand: 17. Dezember 2025

Inhaltsverzeichnis

1 Introduction

Electrical components can be divided into different categories depending on their function and mode of operation. There are resistors, capacitors, coils, but also diodes and transistors, which will be discussed in a later chapter. Electrical components must be supplied with energy via a source (current or voltage source) in order to perform their desired function. Figure 1.1 shows various components.

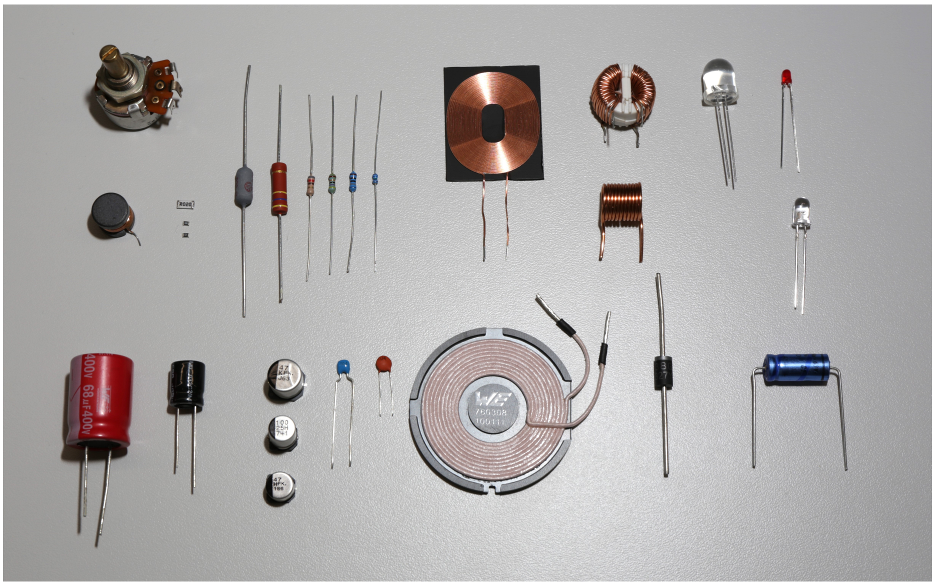


Figure 1.1: **Various electrical components.** Shows electrical components used in circuits, such as resistors, capacitors, coils and semiconductor devices. Each component has specific properties and functions for controlling current and voltage in an electrical network.

Each of these components uses the basic principles of electrical engineering to perform specific functions in an electrical circuit. By combining different components, complex circuits can be created that are capable of performing demanding tasks, such as processing data in a computer or regulating the speed of an electric motor.

The transition from fundamental physical concepts to electrical components is therefore a step from theory to practice, enabling us to apply our theoretical knowledge in real-world applications. This chapter introduces the three fundamental electrical properties of components, as well as voltage and current sources.

2 Conductivity and resistance

Conductivity, or conductance G, describes, as the name implies, the ability of a material to conduct electrical current. The higher the value, the better a material conducts, i.e., the easier it is for charge carriers to move within the material. This property is based on the experiments of German experimental physicist Georg Simon Ohm in the early 19th century, in which he discovered that the current is proportional to the applied voltage. The reciprocal of the conductance is the more commonly used ohmic resistance R. The conductance G has the unit Siemens (S), while the ohmic resistance has the unit ohm. (Ω).

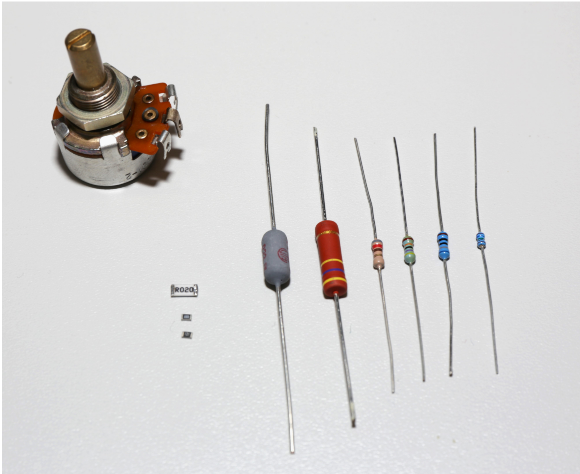


Figure 2.1: **Unterschiedliche ausführungen von Widerständen.** These include wirewound resistors, potentiometers and SMD resistors. The design and material determine the electrical properties, such as resistance value, load capacity and temperature coefficient.

In circuit design, the physical property of ohmic resistance is used to achieve the desired functions. Figure 2.1 shows different designs of electrical resistors. Regardless of the design, all ohmic resistors are based on the fundamental principles of physics, which will be discussed in more detail below.

Learning objectives: Leitfähigkeit und Widerstand

The Students

- are familiar with the electrical component known as a resistor.
- can explain the different component designs of resistors.
- can perform calculations using the specific resistance ρ or the specific conductance κ.
- can analyse the differences in conductivity of various materials based on their atomic structure and temperature.

2.1 Electrical conductivity

The definition of current density J and current I is known from Module 1. Substituting formulas 2 and 3 into formula 1 yields formula 4:

$$J = \frac{\Delta I}{\Delta A} \tag{2.1}$$

$$I = \frac{\Delta Q}{\Delta t} \tag{2.2}$$

$$\Delta V = \Delta A \cdot \Delta x \Leftrightarrow \Delta A = \frac{\Delta V}{\Delta x} \tag{2.3}$$

$$J = \frac{\Delta Q \cdot \Delta x}{\Delta t \cdot \Delta V} \tag{2.4}$$

2.2 Electrical conductivity

By inserting the drift velocity v_{el} and the space charge density ρ known from Module 1 into Formula 4, Formula 7 for the current density is obtained.

$$v_{el} = \frac{\Delta x}{\Delta t} \tag{2.5}$$

$$\rho = \frac{\Delta x}{\Delta V} \tag{2.6}$$

$$\vec{J} = \rho \cdot \vec{v_{el}} \tag{2.7}$$

Since velocity always has a magnitude and a direction, current density is also a vector.

2.3 Electrical conductivity

The drift velocity $\vec{v_{el}}$ of the free charge carriers is proportional to the electric field $\vec{E}$, but in the opposite direction. The proportionality factor is also referred to as electron mobility μ_e.

$$\vec{v_{el}} = -\mu_e \cdot \vec{E} \tag{2.8}$$

The space charge density ρ is composed of the elementary charge e^- and the charge carrier density n_e. Due to the negative charge of electrons, this results in a negative sign.

$$\rho = -n_e \cdot e^- \tag{2.9}$$

Substituting $\vec{v_{el}}$ and ρ into formula 7 yields the following expression:

$$\vec{J} = (-n_e \cdot e^-) \cdot (-\mu_e \cdot \vec{E}) \tag{2.10}$$

2.4 Electrical conductivity

The minus signs cancel each other out, resulting in the following formula for the current density $\vec{J}$:

$$\vec{J} = n_e \cdot \mu_e \cdot e^- \vec{E} \tag{2.11}$$

The material-dependent components n_e and μ_e as well as the natural constant e^- multiplied together yield the specific conductivity κ:

$$\kappa = n_e \cdot \mu_e \cdot e^- \tag{2.12}$$

In summary, this description results in the description of Ohm's law:

$$\vec{J} = \kappa \cdot \vec{E} \leftrightarrow \vec{E} = \frac{\vec{J}}{\kappa} \tag{2.13}$$

$$U_{12} = \int_{P_1}^{P_2} \vec{E} \mathrm{d}\vec{s} \tag{2.14}$$

Since the electric field runs parallel to the length of the conductor, the scalar product is:

$$U_{12} = \int_{x=0}^{x=l} \vec{E} \mathrm{d}\vec{x} \tag{2.15}$$

By substituting the rearranged formula 13 into $\vec{E}$, the following follows:

$$U_{12} = \int\limits_{0}^{l} \frac{J}{\kappa} \mathrm{d}x \tag{2.16}$$

Since it is a homogeneous material, κ is constant.

$$U_{12} = \frac{1}{\kappa} \cdot \int\limits_{0}^{l} J \mathrm{d}x \tag{2.17}$$

The following formula for current density J is known from Module 1:

$$J = \frac{I}{A} \tag{2.18}$$

Inserting the formula yields the following expression:

$$U_{12} = \frac{1}{\kappa} \cdot \int\limits_{0}^{l} \frac{I}{A} \mathrm{d}x \tag{2.19}$$

Electrical conductivity indicates how well a material can transport charge carriers, i.e. how well electrical current flows through it. Materials with very good conductivity are called conductors (in extreme cases also superconductors), while materials with very poor conductivity are called insulators. Conductive media include metals such as copper, silver, aluminium and gold, which are most commonly used in electrical engineering. Insulators include plastics (polyethylene (PE), polytetrafluoroethylene (PTFE or Teflon), polyvinyl chloride (PVC), etc.), mineral oils, silicones and epoxy resin (e.g. as casting compound) specially manufactured for use as insulators, or even paper. There are also semiconductors such as silicon and germanium, whose conductivity falls between that of conductors and insulators. Their conductivity can be altered by doping and external influences such as temperature and light. These properties are used in components such as diodes or transistors to perform certain functions. The different types and properties of semiconductors are very extensive and will be covered in later modules. For the sake of completeness, it should be mentioned that liquid media such as acids or salt solutions are conductive and are referred to as electrolytes. There are also ionised gases that contain freely moving charge carriers and can conduct electric currents. These conductive gases are called plasmas, which are also not covered in this module.

2.5 Drift behaviour of electrons in conductors

In Module 1, we already learned that electric current flow is the directed movement of free charge carriers. In the case of a conductor, these are the negative charge carriers, i.e. the electrons, which can move freely within the material. The structure of such a material consists of a solid lattice of atomic nuclei and freely moving electrons. Both the mobility of the electrons μ_e and their number depend on the material. In a field-free space, these electrons move randomly in all directions, resulting in a neutral state. Since this is not a directed movement but one that is evenly distributed in all directions, no current flow can be detected. Nevertheless, this undirected, temperature-dependent movement of the charge carriers leads to electrotechnical effects: In sensitive circuits, it is perceived as noise. The free charge carriers are much lighter than the atomic nuclei anchored in the lattice, so that when a moving electron collides with an atomic nucleus, an inelastic collision occurs and the direction of motion of the electron changes. This behaviour can be seen in Figure 2.2.

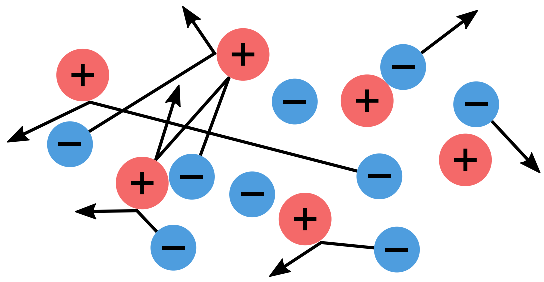


Figure 2.2: **Undirected movement of free electrons.** Shows the free electrons (blue) between positively charged atomic nuclei (red). Such movements are characteristic of metals at rest, where the electrons move randomly, without a preferred direction, as long as no external electric field is applied.

As soon as a conductor is placed in an electric field, the Coulomb force acts on the electrons, causing their random direction of movement to change into a directed movement that points in the opposite direction to the field (Abb.2.3). This directed movement of the charge carriers is referred to as current flow (see Module 1) . However, collisions with the atomic nuclei continue to occur. These ultimately lead to a weakening of the directional motion and thus of the current (see Fig. 2.3). As the atomic nuclei move more rapidly with increasing temperature, the probability of collisions between the electrons and the nuclei increases with increasing temperature.

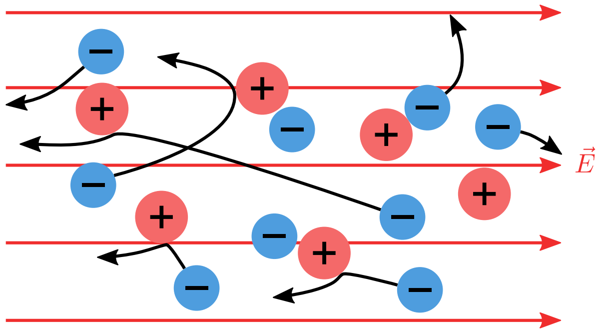


Figure 2.3: **Directed movement of electrons.** Under the influence of an electric field $\vec{E}$, the free electrons in a conductor move in the opposite direction to the field, which explains the electric current flow I.

This behaviour of electrons is also called drift behaviour, which depends heavily on the material. Since the atomic composition depends on the material, the number of free electrons and their mobility μ_e also differ. These two properties are summarised under the term specific conductivity κ. The specific conductivity can thus be used to determine the drift velocity of the electrons in the lattice structure of the conductor. The following section deals with these properties.

2.6 Mathematical relationships of conductivity

The different atomic compositions of the materials result in material-dependent electrical conductivity. The material-specific properties consist of the specific conductivity κ (kappa) or the specific resistance ρ_R (rho), which are inversely proportional to each other, as well as the respective temperature coefficient α, which differs depending on the material.

$$[\kappa] = 1\,\frac{\text{Siemens}}{\text{m}} = 1\,\frac{\text{S}}{\text{m}} = 1\,\frac{1}{\Omega \cdot \text{m}}$$

$$\kappa = \frac{1}{\rho_{\text{R}}} \tag{2.20}$$

$$\rho_{\text{R}} = \rho_{20^\circ\text{C}} \cdot (1 + \alpha(\vartheta - 20^\circ\text{C})) \tag{2.21}$$

The temperature of the conductor is entered for ϑ (theta). This results in $(\vartheta - 20°C)$, which is the temperature difference between the actual temperature of the conductor and the 20°C at which the reference value was determined. If $20°C$ is used for ϑ (theta), α is also omitted and the reference value $\rho_{20^\circ\text{C}}$ remains. The following table shows examples of some materials with their specific properties:

Material	**Specific conductivity** $[\kappa] = \text{S/m}$	**Temperature coefficient** $[\alpha] = 1/\text{K}$
Silver	$6.1 \cdot 10^7$	0.0038
Copper	$5.8 \cdot 10^7$	0.0038
Gold	$4.5 \cdot 10^7$	0.0034
Aluminium	$3.7 \cdot 10^7$	0.004
Iron	$1.0 \cdot 10^7$	0.0065
Graphite	$3 \cdot 10^6$	-0.0002
Silicon (dotiert)	$1 - 10^6$	-0.075
Tap water	$5.0 \cdot 10^{-3}$	-
Air	$4.0 \cdot 10^{-15}$	-

Table 2.1: **Specific conductivities and temperature coefficients of various materials.** Metals such as silver and copper have high conductivity, while insulating materials such as air have very low conductivity.

Equivalent to the specific conductivity κ and the specific resistance ρ_{R}, the corresponding resistance R or conductance G of a body can be determined.

$$[G] = 1\,\text{Siemens} = 1\,\text{S} = 1\,\frac{1}{\Omega}$$

$$G = \frac{1}{R} \tag{2.22}$$

The electrical conductivity or resistance of a conductor is also determined by its geometric properties. Both the length l and the cross-sectional area A contribute to the conductivity or resistance value. Figure 2.4 shows such a conductor with its properties.

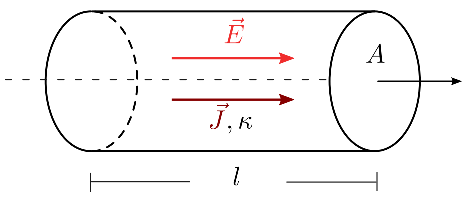


Figure 2.4: **Geometric representation of a resistor.** The electric field $\vec{E}$ generates a current density $\vec{J}$ through the resistance. The relationship between the current density and the electric field is described by Ohm's law $\vec{J} = \kappa \cdot \vec{E}$.

The current density $\vec{J}$ is derived from the electric field $\vec{E}$ and the specific conductivity κ. The current density is independent of geometry. Only when calculating the current I must the length and cross-sectional area be taken into account. The resistance value can be calculated using the following formula:

$$[R] = 1\,\mathrm{Ohm} = 1\,\Omega$$

$$R = \rho_\mathrm{R} \cdot \frac{l}{A} \tag{2.23}$$

$$\begin{aligned} \rho_\mathrm{R} &: \text{spezifischer Widerstand des Materials } (\Omega\mathrm{m}) \\ l &: \text{Länge des Materials } (\mathrm{m}) \\ A &: \text{Querschnittsfläche des Materials } (\mathrm{m}^2) \end{aligned}$$

Introduction to the representation of circuits

In electrical engineering, so-called circuit diagrams are used to represent electrical networks. These circuit diagrams contain circuit symbols, which are used to represent electrical properties or entire components. Figure 2.5 shows an example of such a circuit diagram.

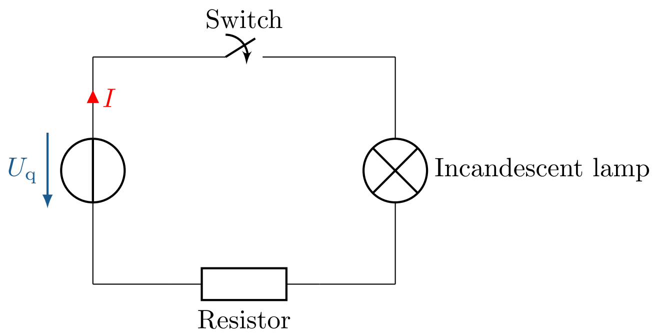


Figure 2.5: **Circuit diagram of a series connection.** Consisting of a DC voltage source, a switch, a load and a resistor.

The circuit diagram consists of four elements connected by lines. It represents a series connection consisting of a voltage source, a switch, an incandescent lamp and a resistor. The resistor represents the line resistance of the entire line, while the connecting lines represent ideal connections without any other properties by definition.

Furthermore, a voltage U_q and the current I are shown, but the current can only flow when the switch is closed. The three basic electrical properties and their circuit symbols are discussed in more detail below.

2.7 Resistance as a component

Using the method for calculating resistance described in the previous section, the resistance value of electrical cables or other conductive objects can be determined. Once a resistance value is known, it is assigned to a circuit symbol in electrical circuit diagrams. This allows different components and their properties to be connected to each other in order to determine their behaviour. Figure 2.6 shows two common circuit symbols for resistance.

Figure 2.6: **Circuit element for resistance.** European (left) and American (right).

As soon as a DC voltage source is connected to a resistor R, the current and voltage behave linearly. Figure 2.7 shows the current I_R flowing through the resistor, as well as the voltage drop U_R.

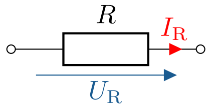


Figure 2.7: **Ideal resistance.** Shows the voltage drop across the resistor and the current flowing through it.

The circuit symbol for resistance is always considered to be an ideal resistor in circuits. If temperature dependence is neglected, this is largely true in the case of direct current (except for switching operations). In reality, there are many situations in which a real resistor cannot exhibit linear behaviour. Different types of sources are discussed in sections 1.2 on voltage and current sources. To understand Figure 2.8, it is advisable to read sections 1.3 and 1.4 first.

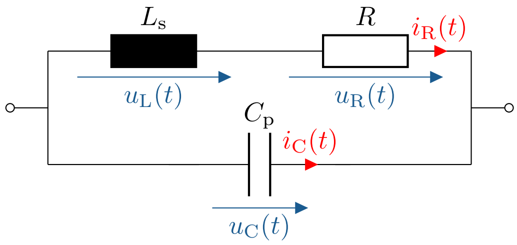
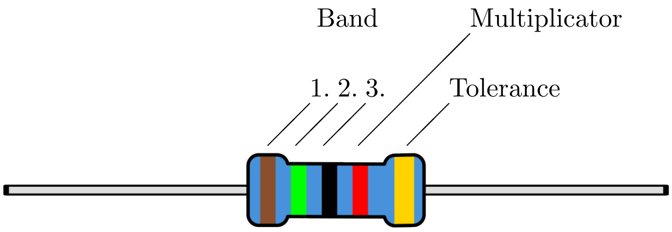


Figure 2.8: **Equivalent circuit diagram of a real resistor.** The ideal resistance R is supplemented by a series inductance L_s and a parallel capacitor C_p to account for parasitic effects.

Based on the design of a resistor and the frequency at which it is operated, so-called parasitic effects become noticeable. Figure 2.8 shows the equivalent circuit diagram of a real resistor. If this is operated at higher frequencies, the parasitic properties (parasitics) can predominate and lead to completely different behaviour. More on this in later modules. However, complete equivalent circuit diagrams are always assumed in the following. If a resistor is only represented by its circuit symbol, it should be assumed that this is a complete description. This also applies to all components derived in the following.

In electrical engineering, resistors, also known as fixed resistors, are used in circuit design. One of the best-known types are THT (through-hole technology) resistors. Although this type has been replaced by SMD (surface-mount device) components in many applications, there are still cases where they make sense. Examples include prototyping, DIY projects, power applications, or printed circuit boards with high mechanical requirements.

THT resistors are marked with coloured rings that provide information about the resistance value and tolerance. Figure 2.9 shows a THT resistor with the corresponding reading scheme and a table for reading the values:

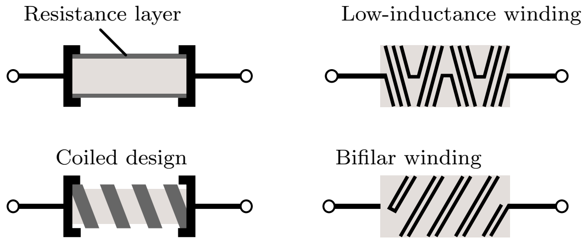


$$1-5-0-2 = 150 \cdot 100\,\Omega = 15\,\mathrm{k}\Omega \pm 5\%$$

Colour	1. 2. 3. Band	Multiplicator	Tolerance
Black	0	$1\,\Omega$	-
Brown	1	$10\,\Omega$	$\pm 1\,\%$
Red	2	$100\,\Omega$	$\pm 2\,\%$
Orange	3	$1\,\mathrm{k}\Omega$	-
Yellow	4	$10\,\mathrm{k}\Omega$	-
Green	5	$100\,\mathrm{k}\Omega$	$\pm 0.5\,\%$
Blue	6	$1\,\mathrm{M}\Omega$	$\pm 0.25\,\%$
Violette	7	$10\,\mathrm{M}\Omega$	$\pm 0.1\,\%$
Grey	8	-	$\pm 0.05\,\%$
White	9	-	-
Gold	-	$0.1\,\Omega$	$\pm 5\,\%$
Silver	-	$0.01\,\Omega$	$\pm 10\,\%$

Figure 2.9: **Resistance colour coding.** Shows the colour codes for determining the resistance value and tolerance of a resistor.

THT resistors are available in various designs, each with their own advantages and disadvantages. Figure 2.10 shows four ways in which these can be implemented.



(a) Layer resistors

(b) Wirewires

Figure 2.10: **Various types of through-hole (THT) resistors.**

Key point:

The resistance value of a material depends on its geometric and specific properties.

2.8 PTC and NTC

This section deals with a special category of resistors that are particularly useful in temperature-dependent applications: PTC and NTC resistors. The names refer to their positive and negative temperature coefficients α, which were already introduced in section 1.1.3. These special resistors, also known as thermistors, differ from conventional resistors in that their resistance value is highly dependent on temperature. Figure 2.11 shows the opposite behaviour of both types of resistance.

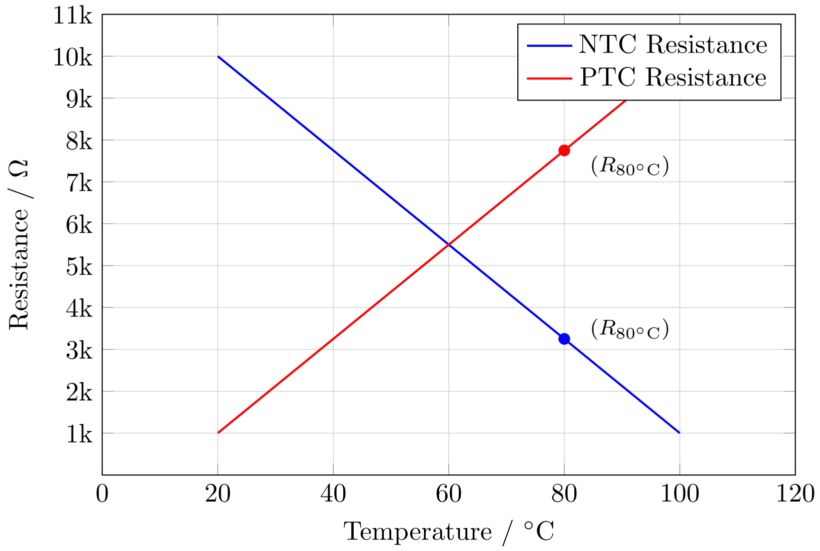


Figure 2.11: **Diagram of a linear NTC and a PTC resistor over temperature.** With an NTC, the resistance decreases as the temperature rises, while with a PTC, the resistance value increases as the temperature rises.

PTC resistors have a positive temperature coefficient α, which can be found in Table 1 in Section 1.1.3. Highly conductive materials have a positive α, which means that their resistance increases at higher temperatures. This makes them more conductive when cold, which is why they are also called 'cold conductors'. Figure 2.12 shows the same PTC resistor at different temperatures.

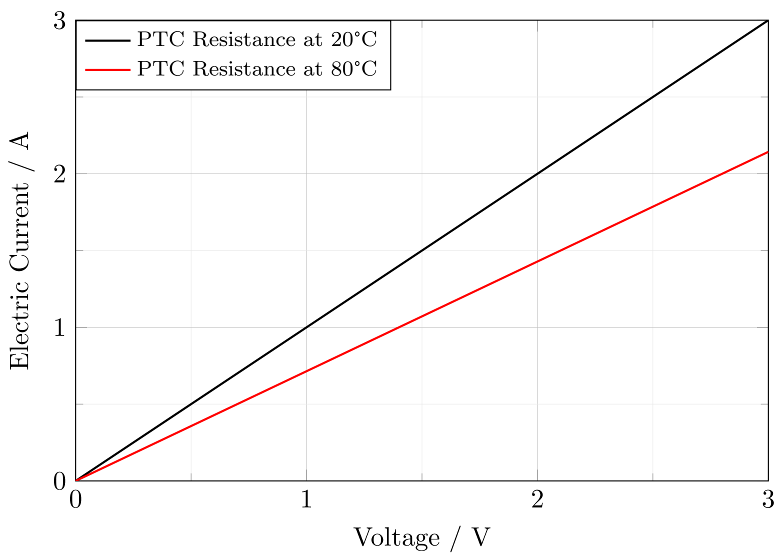


Figure 2.12: Diagram of a PTC resistor at 20°C and at 80°C

The following diagram shows the heating of a PTC resistor when the current flow increases. The more current flows through a PTC resistor, the more it heats up.

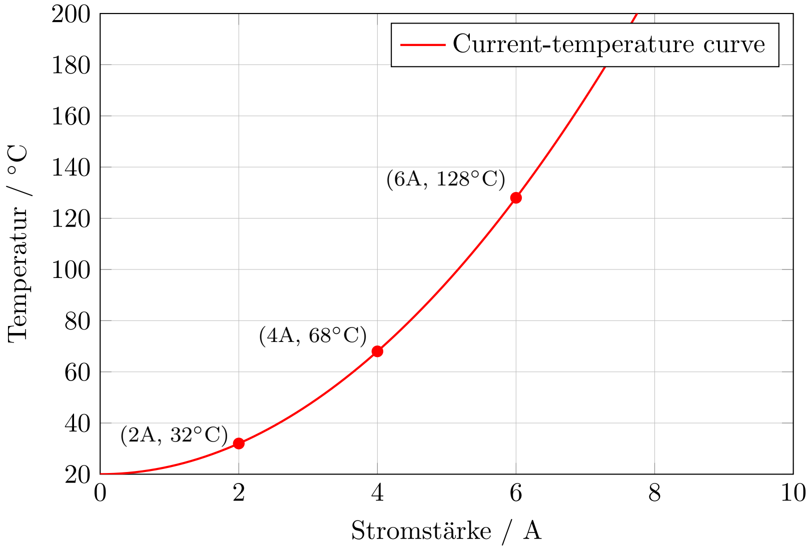


Figure 2.13: **Diagram showing the temperature of a resistor as a function of current.** Shows how the temperature of a resistor rises with increasing current. The resistance value may also change, depending on the temperature dependence of the resistor.

An NTC resistor exhibits the exact opposite behaviour due to its negative temperature coefficient. Typical materials with this behaviour are graphite or silicon. Since these resistors conduct better at higher temperatures, they are also called thermistors. Figure 2.14 shows that the resistance has a lower resistance value at a higher temperature.

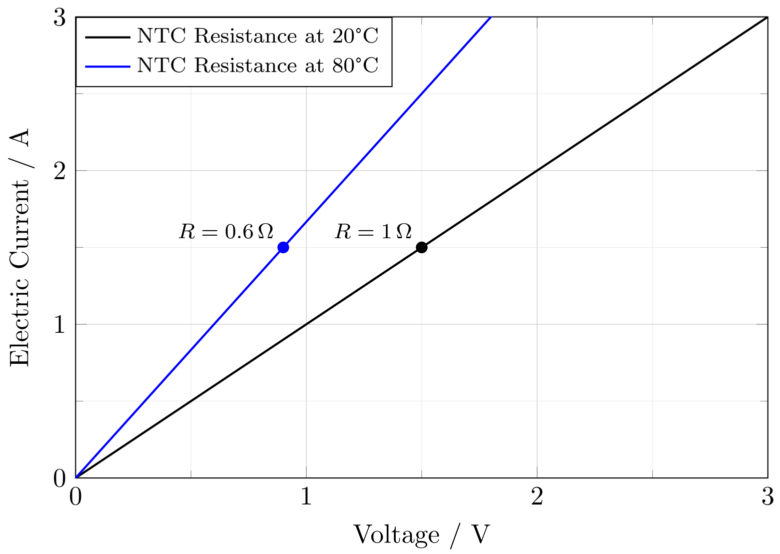


Figure 2.14: Diagram of an NTC resistor at 20 ˚C and at 80 ˚CC

3 Voltage and current source

The operation of electrical devices is usually associated with the conversion of electrical power. In Module 2 on the topic of energy and power, we already learned that power corresponds to the amount of work done per unit of time and that this is equivalent to energy conversion. In order to operate electrical devices, in which energy is always converted, energy must be supplied. Energy can only be converted, not generated. Nevertheless, in electrical engineering, the supply of energy to the circuits under consideration is described with the aid of current and voltage sources. This does not take into account the fact that this involves the conversion of, for example, mechanical or chemical energy into electrical energy. However, this is irrelevant for the consideration of the circuit. The following section describes current and voltage sources and explains the differences between them.

Learning objectives: Spannungs- und Stromquelle

The Students can

- distinguish between voltage and current sources and know their circuit symbols.
- model real voltage and current sources.
- name examples of different sources.

3.1 Voltage sources

The most common type of energy source for electrical systems is the voltage source. Ideally, it is characterised by the fact that it outputs a constant voltage regardless of the connected load. Figure 3.1 shows the circuit symbol of an ideal voltage source with a constant output voltage U_q.

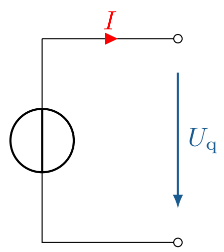


Figure 3.1: **Ideal general voltage source.** Always supplies a constant voltage U_q, regardless of the load, and automatically adjusts the current to maintain the constant voltage.

In Figure 3.1, the lines lead to nothing, which is referred to as open terminals. Since the circuit is not closed, no current flows. If the terminals were short-circuited in this case, the source would attempt to maintain the voltage and an infinitely high current would flow. Why this is so dangerous is explained in section 3.2 using a real voltage source as an example.

3.2 Modelling real voltage sources

Ideal components are always used in circuit diagrams, i.e. the graphical representation of electrical circuits. In order to be able to model the real behaviour of components correctly and thus replicate the real circuit behaviour as accurately as possible, the actual behaviour of the component must be replicated with the aid of ideal components. The following describes how the behaviour of a real voltage source can be replicated using ideal components. The technical implementation of any voltage source consists of physical components, all of which have specific resistance values. All these properties together result in the so-called internal resistance R_i of a source. In the case of a real voltage source, the internal resistance is in series with the ideal voltage source. Figure 3.2 shows a modelled real voltage source.

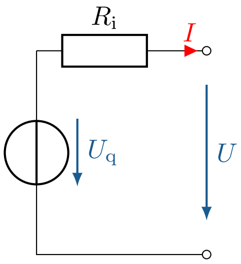


Figure 3.2: **Real voltage source with internal resistance.** The voltage U_q is influenced by the internal resistance R_i and is therefore lower at the output terminals.

The circuit diagram shows that both U_q and U point downwards. This is due to the counting arrow system, which is discussed in more detail in Module 4. At this point, it is sufficient to know that the generator counting arrow and the consumer counting arrow are opposite, since the sum of both arrows must be zero. If a consumer is now connected, the current flows through the internal resistance R_i through the consumer. This means that power is also converted in the internal resistance.

If we consider the behaviour of short-circuited output terminals in a real-world scenario, the entire power delivered by the source is converted solely by the internal resistance of the voltage source. The power can be calculated using $P = \frac{U^2}{R_\mathrm{i}}$. As a rule, the source is not designed for this, which would result in the energy being converted into a great deal of heat energy and lead to the destruction of the source.

If a load with a resistance R_L is connected to the output of the voltage source, a current I flows through the resistors R_i and R_L. This can be calculated according to Ohm's law and the summary of resistors presented in the following module as $I = \frac{U_\mathrm{q}}{(R_\mathrm{i}+R_\mathrm{L})}$. This current causes a voltage $U_{\mathrm{R_i}}$ to drop across the internal resistance R_i. In Module 2, it was already derived that the sum of all voltages within a closed circuit must be zero. In Module 4, the so-called mesh rule is derived from this, which can be used to calculate the output voltage U as

$$U = U_\mathrm{q} - U_{\mathrm{R_i}} = U_\mathrm{q} - I \cdot R_\mathrm{i} = U_\mathrm{q} - U_\mathrm{q} \cdot \frac{R_\mathrm{i}}{R_\mathrm{i} + R_\mathrm{L}} \tag{3.1}$$

The smaller the internal resistance R_i, the less the output voltage U of a loaded voltage source differs from the source voltage U_q. An internal resistance R_i that is as small as possible in comparison to the load resistance R_L should therefore be aimed for in order to achieve the most ideal behaviour possible. This can be achieved, among other things, by using the largest possible conductor cross-sections within the source.

If the real voltage source, modelled with U_q and R_i, is operated in idle mode, the output voltage U is equal to the source voltage U_q, since no current flows through the internal resistance and no voltage can drop there.

Examples of real voltage sources include batteries and accumulators. As described above, the aim is to achieve the lowest possible internal resistance. A 12 V starter battery for cars has an internal resistance of less than 10 mΩ. A domestic power socket can also be regarded as a voltage source with a source voltage of 230 V and low internal resistance.

Key point:

- A voltage source must never be short-circuited!
- With voltage sources, the current always adjusts itself.

3.3 Current sources

Compared to voltage sources, current sources are much less common. They are characterised by the fact that they supply a constant current and the voltage adjusts accordingly. Figure 3.3 shows the circuit symbol of an ideal current source with a constant output current. I_q.

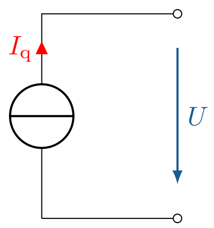


Figure 3.3: **Ideal general power source.** Always supplies a constant current I_q, regardless of the load, and automatically adjusts the voltage to maintain the constant current.

The case of an idle current source shown here leads, similarly to the short-circuited voltage source, to an infinitely high power output from the source and must therefore be avoided.

3.4 Modelling real current sources

Just as with voltage sources, there are no ideal current sources in reality. In order to describe the real behaviour of a current source, it must first be modelled. Figure 3.4 shows such a modelled current source.

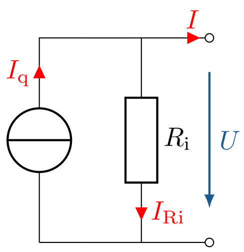


Figure 3.4: **Real current source with internal resistance.** The current I_q is influenced by the internal resistance R_i and is lower at the output terminals.

The model of the real current source also consists of the internal resistance R_i and the ideal current source, where the difference to the voltage source is that the internal resistance is placed in parallel to the ideal current source. Here, too, the actual output current is smaller than the source current I_q. Module 4 introduces the node rule. This states that the sum of all currents in a node (branch point) must be zero, so that the output current I is given by

$$I = I_\mathrm{q} - I_{\mathrm{R_i}} = I_\mathrm{q} - \frac{U}{R_\mathrm{i}} \tag{3.2}$$

The greater the internal resistance R_i, the smaller the current I_{R_i} and the smaller the difference between the output current I and the source current I_q.

Contrary to the circuit diagram shown, a current source must never be operated with open terminals, as, similar to the case of a voltage source, the entire power would be converted in the internal resistance, which would lead to high thermal stress and ultimately to the destruction of the current source. Examples of real current sources are solar cells in the corresponding operating state or a laboratory power supply in current limiting mode.

Key point:

- A power source must never be operated in idle mode!
- With power sources, the voltage always adjusts itself.

3.5 Conversion of voltage and current sources

Voltage and current sources can be converted into each other with regard to their terminal behaviour. The prerequisite for this is that both the short-circuit current I_K and the open-circuit voltage U_L are the same for both sources. Figure 3.5 shows the conversion of the circuit diagrams into each other. Further information on conversion and power matching will follow in Module 4.

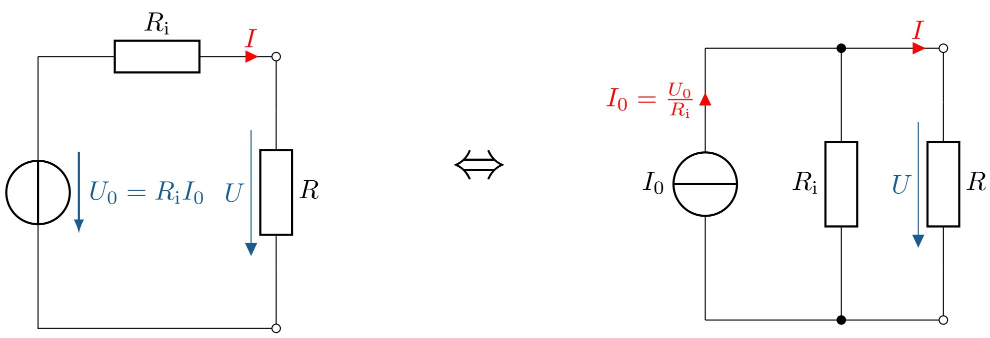


Figure 3.5: **Conversion of a DC voltage source into a current source.** The internal resistance of the voltage source becomes the parallel resistance of the current source, whereby the resulting current source supplies a constant current determined by the resistance.

3.6 Various voltage and current sources

The previously introduced circuit symbols for voltage and current sources are universal and do not make any specific statements about the signal form of the output variable. In addition to the distinction between voltage and current sources, a fundamental distinction is also made between direct current (DC) and alternating current (AC).

The most common form for operating electronic devices is the direct voltage source. It continuously ensures that charge carriers flow from the negative to the positive pole, which generates a constant electric field from the positive to the negative pole.

Alternating current (AC) sources, on the other hand, typically have a sinusoidal output voltage or a sinusoidal output current. In the past, high-voltage lines with alternating current have become the standard for transporting energy over long distances, as they were technically simpler and more cost-effective to implement. The most significant advantage of alternating current is the simple transformation of the voltage. It is also practical because there is no need to pay attention to polarity when connecting consumers.

However, the first implementations of energy transport in the form of direct current voltage are now also available. The use of direct current (DC) for long transmission distances, such as in the HVDC (High Voltage Direct Current) lines from the North Sea to southern Germany, offers a number of advantages: Direct current is more efficient because it does not generate reactive power, which means that there are no additional losses due to cable capacity. This keeps the current amplitude lower, resulting in lower transmission losses and making DC particularly attractive for very long distances. Figure 3.6 shows the circuit symbols for the various sources.

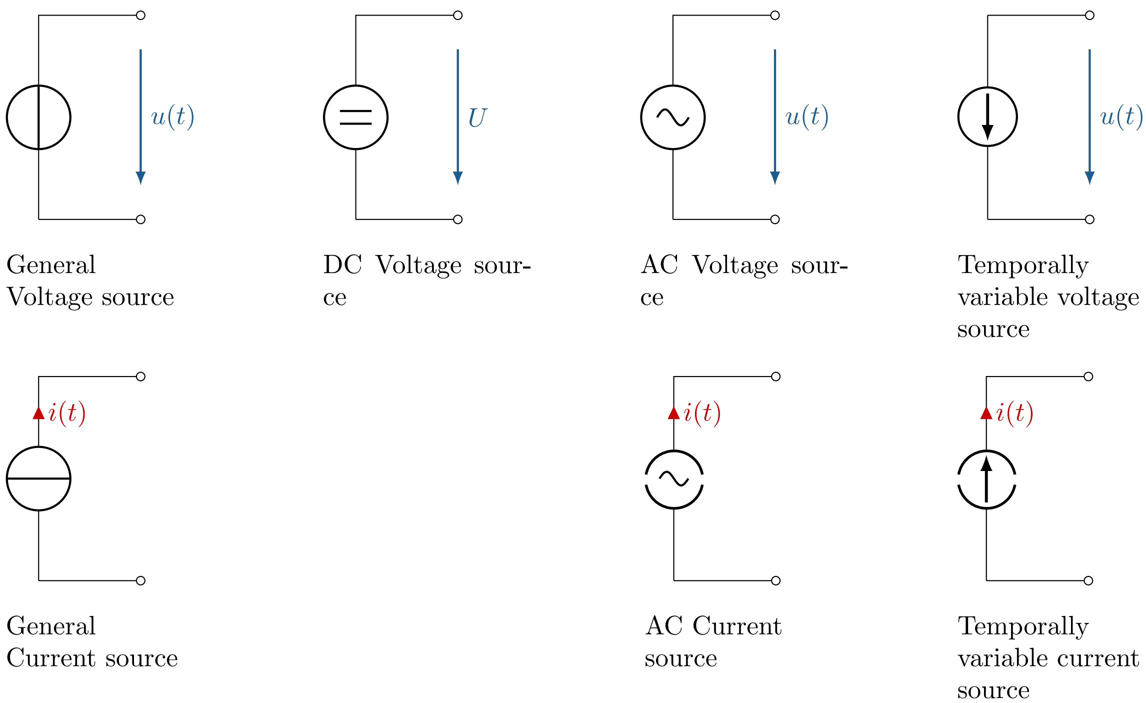


Figure 3.6: **Various voltage and current sources.**

It should be noted that AC voltage and current sources have a sinusoidal output voltage or sinusoidal output current, whereas sources that can be varied arbitrarily over time can have any shape. One example of this is the square wave signal, which could originate from switching a DC voltage source on and off. Such sources are used, for example, in mobile applications of the widely used type of electric motors, which are also called synchronous or asynchronous machines. These motors are operated in mobile applications by a DC voltage source, whose signal shape is converted (pulsed) into the desired square wave signal with the aid of power electronics and a sinusoidal current is generated in the motor from the pulsed voltage signal via the coil windings. More on this in the module on electric machines. Another example of signals that can occur at any time is digital information transmission in communications engineering, where a one corresponds to a high signal and a zero to a low signal, which also resembles a square wave signal.

4 Capacity and capacitor

In electrical engineering, there are three basic components that can be used to describe the electrical behaviour of components. In addition to the ohmic resistance discussed above, there are also capacitance and inductance. This section deals with the property of capacitance and the associated component, the capacitor. The following section deals with inductance and the associated component, the coil.

Learning objectives: Kapazität und Kondensator

The Students

- can differentiate between capacity and capacitor.
- are familiar with the most important parameters relating to capacitors and capacitance.
- can calculate the capacity of a capacitor.

4.1 Electrical capacity C

Electrical capacity is the ability of an object to store electrical energy in the form of an electric field. This property occurs when electrical charge carriers are unevenly distributed and an electric field forms between the areas. The symbol for capacity is C (capacity), while the unit is given in farads (F).

The simplest and most common example used to explain capacitance is the plate capacitor. In this type of capacitor, two conductive plates are positioned opposite each other, each of which is charged with a different potential. Figure 4.1 shows a capacitor with an idealised field distribution.

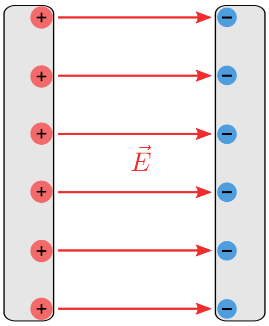


Figure 4.1: **Ideal plate capacitor with idealised field distribution.** The capacitor stores electrical energy in an electric field, which is ideally represented between the plates, and has a constant capacitance C that depends only on the area of the plates and the distance between them.

Due to the different potentials on the plates, an electric field forms from the higher to the lower charged plate. Ideally, it is assumed that no field forms outside the capacitor plates. In reality, charge carriers also accumulate at the side edges of the plates, leading to more complex field distributions. Figure 4.2 shows a diagram of such a real plate capacitor.

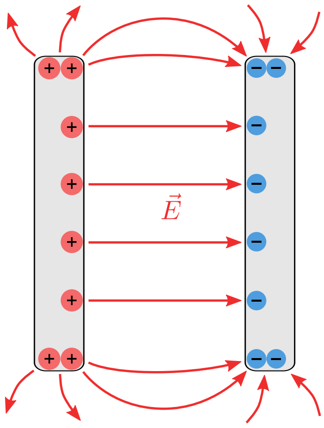


Figure 4.2: **Real capacitor with real field distribution.** In contrast to the ideal capacitor, edge fields and losses occur, which influence the actual capacity and electrical behaviour.

In many cases, however, the real behaviour can be neglected, which greatly simplifies the calculations. For this reason, only idealised capacitors will be considered in the following.

Capacitance is defined as the ratio of the stored electrical charge Q to the applied voltage U. The higher the capacitance of a capacitor, the more charge it can store at a given voltage.

$$[C] = 1\,\mathrm{Farad} = 1\,\mathrm{F} = 1\,\frac{\mathrm{C}}{\mathrm{V}} = 1\,\frac{\mathrm{As}}{\mathrm{V}}$$

$$C = \frac{Q}{U} \tag{4.1}$$

$$Q = C \cdot U \quad \Rightarrow \quad C = \frac{Q}{U} = \frac{\varepsilon \cdot \iint \vec{E} \cdot \mathrm{d}\vec{A}}{\int \vec{E}\mathrm{d}\vec{s}} \tag{4.2}$$

4.2 Der Kondensator als Bauelement

The properties of capacitance are deliberately exploited in various areas of electrical engineering. These properties are utilised with the aid of capacitors. They are available in various designs. Figure 4.3 shows some examples, ranging from THT to SMD components.

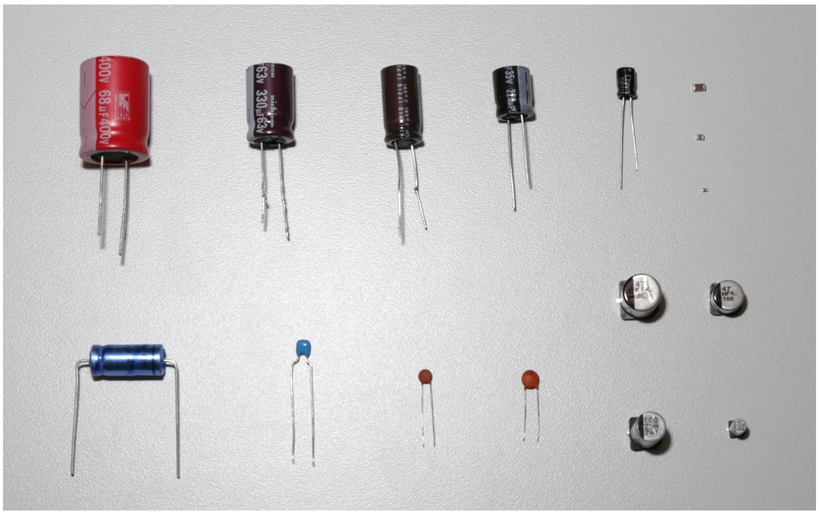


Figure 4.3: **Various types and designs of capacitors.** These include ceramic and electrolytic capacitors. These vary in size, capacity and voltage to suit different applications.

However, they are also available in significantly smaller or larger dimensions. In high-frequency technology, for example, capacitance is achieved solely through the design of the printed circuit boards, whilst capacitors for industrial or energy technology applications can be several centimetres to metres in size.

What all capacitors have in common is their ability to utilise the property of capacitance, as shown in the following circuit diagram 4.4. However, this circuit diagram only represents the idealised, usable capacitance.

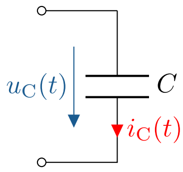


Figure 4.4: **Switching element of an ideal capacitor.** Shows the voltage drop across the capacitor and the current flow through it.

For a realistic simulation of the circuit, it is important to describe the capacitor as a component in its entirety. This is done with the aid of an equivalent circuit diagram, which also simulates the unwanted, i.e. parasitic, properties of the component. Figure 4.5 shows such an equivalent circuit diagram.

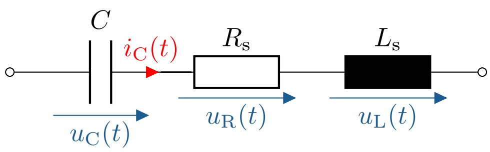


Figure 4.5: **Equivalent circuit diagram of a real capacitor.** The ideal capacitor C is supplemented by a series inductance L_{s} and a series resistance R_{s} to account for parasitic effects.

The equivalent circuit diagram of the capacitor takes into account not only the usable capacitance, but also the parasitic properties that occur in real capacitors. These parasitic properties arise, for example, from the inductance of the connections and the ohmic losses in the material. The modelled equivalent circuit allows these effects to be taken into account in the circuit design and the switching behaviour to be better predicted.

Key point:

The capacitor is a desperate attempt to replicate a capacitance.

4.3 The dielectric

As the name implies, a dielectric is a dielectric material, i.e. one that does not conduct electricity. Since every material consists of atoms and all atomic nuclei are surrounded by electrons (electron shell), even a material that does not conduct electricity has an effect on the behaviour of the electric field. As soon as such a material is inserted between the plates of a capacitor, the properties of the capacitor also change. In Figure 4.6, the interrupted arrow of the electric field shows the different behaviour. The electric flux density D is an auxiliary quantity that is independent of the material and will be discussed in section 1.3.4.

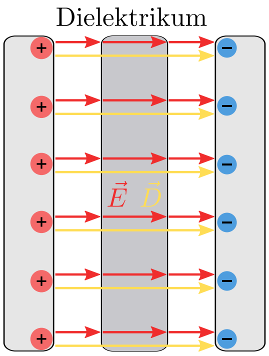


Figure 4.6: **Ideal capacitor with dielectric.** The capacitor consists of two plates with a dielectric between them. The dielectric is a non-conductive material that reduces the electric field strength and increases the capacitance of the capacitor by increasing the relative permittivity.

The properties of the dielectric are material-specific and are quantified by the dielectric constant ε_r. Table 4.1 lists examples of some dielectrics and their ε_r:

Material	$\varepsilon_\mathbf{r}$
Vacuum	1
Air (at STP)	1.0006
Plastic (PE)	2.25-2.3
Glass	3-10
Ceramics	5-300
Silicon	11.7
Tantalum oxide	25-30
Water	81

Table 4.1: **Relative permittivity of various materials.** This dimensionless number indicates how strongly the electric field in a material is affected compared to a vacuum. It is a measure of the material's ability to store electrical charge. Materials with a higher dielectric constant increase the capacity of a capacitor because they reduce the electric field strength inside it.

In order to fully describe the material properties of the dielectric, the electric field constant ε_0 is required in addition to the dielectric constant. This refers to the dielectric behaviour of the electric field in a vacuum. While the dielectric constant ε_r is dimensionless, the electric field constant has the unit $\frac{\mathrm{As}}{\mathrm{Vm}}$. It is multiplied by the relative dielectric constant, which together gives the dielectric constant ε.

$$[\boldsymbol{\varepsilon}] = 1\,\frac{\mathrm{Farad}}{\mathrm{m}} = 1\,\frac{\mathrm{F}}{\mathrm{m}} = 1\,\frac{\mathrm{As}}{\mathrm{Vm}}$$

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_0 \cdot \boldsymbol{\varepsilon}_\mathrm{r} \tag{4.3}$$

$\boldsymbol{\varepsilon}$: Dielectric constant

$\boldsymbol{\varepsilon}_0$: Electric field constant $(8.85421878 \cdot 10^{-12})$

$\boldsymbol{\varepsilon}_\mathrm{r}$: Relative permittivity

With the help of these material properties of the dielectric and the dimensions and distance of the capacitor plates, the capacitance of the capacitor can be calculated as follows:

$$[C] = 1\,\mathrm{F}$$

$$C = \frac{\boldsymbol{\varepsilon} \cdot A}{d} \tag{4.4}$$

A : Cross-sectional area of the material (m^2)

d : Distance between electrodes (m)

4.4 The electric flux density

The electric flux density $\vec{D}$, also known as displacement density, is proportional to the electric field $\vec{E}$. Its advantage is that it is independent of the material, which brings mathematical advantages.

$$\vec{D} = \boldsymbol{\varepsilon} \cdot \vec{E} + P \tag{4.5}$$

In addition to the dielectric constant ε, the polarisation of the medium P is added. It describes the polarisation behaviour that occurs during the switch-on process. In static cases, the polarisation P can often be neglected, which simplifies the equation as follows.

$$[D] = 1\,\frac{\text{Coulomb}}{\text{m}^2} = 1\,\frac{\text{C}}{\text{m}^2} = 1\,\frac{\text{As}}{\text{m}^2}$$

$$\vec{D} = \boldsymbol{\varepsilon} \cdot \vec{E} \tag{4.6}$$

Einführung in die zeitabhängige Vorgänge

It is known that electric fields in conductive materials cause a current I and a voltage U. However, the current I and the voltage U only describe the fields integrally, meaning they are considered to be constant over time. In contrast to resistance R, there are components (capacitors and coils) that exhibit time-varying behaviour. Ohmic resistance R alone is therefore not sufficient to describe the conditions in networks with time-varying processes. The time-dependent variables $u(t)$ and $i(t)$ are used to describe this time-dependent behaviour.

The derivation of the time dependencies can be shown mathematically as follows. The basic equation of the capacitor is already known from formula 4.1:

$$Q = C \cdot U \tag{4.7}$$

To describe the temporal behaviour of the capacitor, the change in charge over time is considered.

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = C \cdot \frac{\mathrm{d}U}{\mathrm{d}t} \tag{4.8}$$

$\frac{\mathrm{d}Q}{\mathrm{d}t}$ describes the rate of change of charge, which, according to the fundamental law of electrical engineering, corresponds to the current $i(t)$ flowing into or out of the capacitor.

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = i(t) \tag{4.9}$$

If the constant charge Q in formula 4.7 is replaced by the time-varying quantity $i(t)$, the voltage also becomes time-dependent.

$$i_\mathrm{c}(t) = C \cdot \frac{\mathrm{d}u_\mathrm{c}(t)}{\mathrm{d}t} \tag{4.10}$$

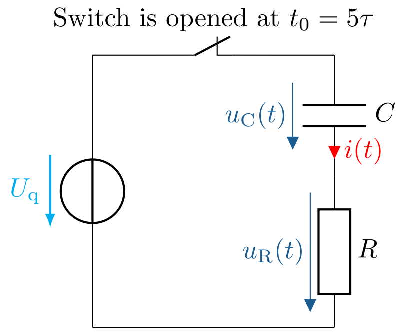
By mathematical conversion to $u_\mathrm{c}(t)$, the voltage can be represented as a function of the current and the capacitance:

$$u_\mathrm{c}(t) = \frac{1}{C} \cdot \int i_\mathrm{c}(t)\mathrm{d}t \tag{4.11}$$

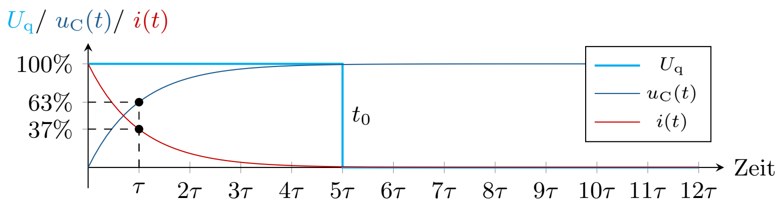
It follows directly from equation 4.10 that the voltage across a capacitor cannot change abruptly, as this would require an infinitely high current.

4.5 Switching behaviour of a capacitor

The switching behaviour describes the behaviour of a component when the applied voltage changes. Depending on whether a voltage source is switched on or off, the current and voltage at the component behave according to its characteristics. To analyse the temporal behaviour of components, a circuit diagram is first required, which visualises the circuit to be analysed. Figure 4.7a shows a circuit diagram of a series connection of a capacitance C with a resistance R, which can be supplied with a voltage U_{q} via a switch.



(a) Representation of the series connection consisting of a capacitance C, a resistor R and a switch.



(b) Voltage and current curve over capacity at a DC voltage that is switched on at time $t = 0$ and switched off at time t_0.

Figure 4.7: **The voltage and current curve over capacity.** The diagram shows the reaction of the capacitance to the sudden change in voltage and illustrates the typical charging behaviour.

The diagram 4.7b shows that despite the applied DC voltage U_{q}, the voltage $u_{\mathrm{c}}(t)$ does not change simultaneously, but exhibits inertia. This time-varying behaviour is expressed in the circuit diagram 4.7a by the time-dependent variables $u(t)$ and $i(t)$.

In this example, the switch is closed at the start and is opened at $t_0 = 5\tau$. The capacitor is not charged at the start. The diagram shows that, despite the applied voltage U_q, the voltage across the capacitance $u_C(t)$ does not change abruptly.

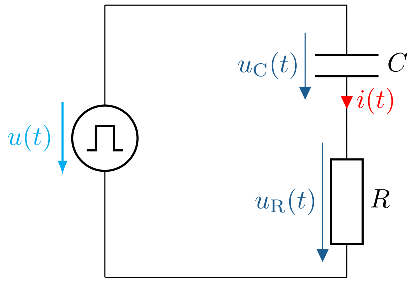
This switch-on behaviour results from the charging of the capacitance. To do this, the charge carriers must accumulate on the capacitor plates. This continues until the voltage between the plates corresponds to the voltage applied by the voltage source. Since the charge carriers required for charging must migrate into the capacitor plates, a corresponding current $i_{\mathrm{C}}(t)$ flows during this process.

In the case of an ideally conductive circuit without ohmic resistance, the charge carriers would accumulate infinitely quickly on the capacitor plates, which would correspond to an infinitely high current. In reality, the capacitor or the lead to the capacitor has ohmic resistance, which slows down the charge carriers on their way to the capacitor plates, thus reducing the electric current. This results in the maximum possible current $I_{\max}$.

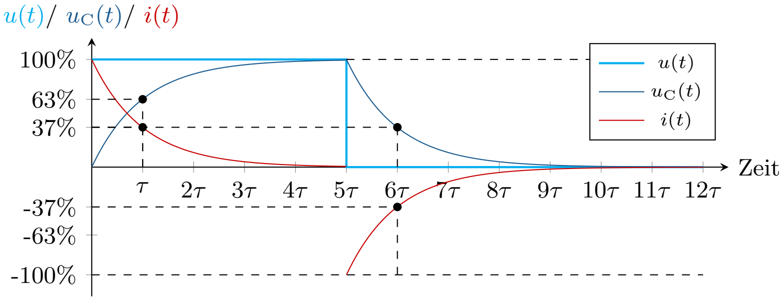
$$I_{\max} = \frac{U_{\mathrm{q}}}{R} \tag{4.12}$$

As soon as the switch is opened, the voltage across the capacitance is maintained. This means that the energy supplied remains in the capacitance and no longer changes in the steady state. Accordingly, no current flows when the capacitance is charged, which is equivalent to no-load operation in a steady-state DC network.

With the help of the two formulas 4.10 and 4.11, the behaviour of current and voltage at a capacitor is fully described, which now enables various calculations. Another example is the switching behaviour of a capacitor when a square-wave voltage is applied. The difference is that, unlike in example 4.7a, a constant voltage source is not switched via a switch, but rather the voltage source itself determines the input voltage.



(a) Representation of the series connection consisting of a capacitance C and a resistance R at a rectangular voltage source.



(b) Voltage and current curve over capacity with a rectangular voltage.

Figure 4.8: **Voltage and current curve over capacity.** The diagram shows the response of the capacitance to a sudden change in voltage and illustrates the typical charging and discharging behaviour of the capacitor.

In contrast to the switch, the voltage source in example 4.8a attempts to pull the system voltage to 0 from the point in time $t_0 = 5\tau$. This leads to the capacitance being discharged. This results in an opposite current flow $i_C(t)$, as the charge carriers move away from the capacitor plates in order to return to a neutral state.

Example 4.1: Capacity calculation

- Charge on plates $Q = \sigma \cdot A = \varepsilon \cdot E \cdot A$
- Calculate capacitance with $C = \frac{\varepsilon \cdot A}{d}$
- Stored energy $W = \frac{1}{2} \cdot C \cdot V^2$
- Calculate the capacitance of two capacitors with different ε_r over $\vec{D}$

5 Inductance and coil

The counterpart to the capacitor is the coil, or more precisely: the counterpart to capacitance C is inductance L. Inductance is also an energy storage device, but the difference lies in the type of fields used. While capacitance stores energy in the form of an electric field, inductance does so via the magnetic field.

Learning objectives: Induktivität und Spule

The Students

- can differentiate between inductance and coils.
- know the most important parameters relating to inductance and coils.
- can calculate the inductance of a coil.

5.1 The inductance L

Inductance is the ability of an object to store electrical energy in the form of a magnetic field. This property occurs when an electric current flows through a conductor, around which a magnetic field forms. The symbol for inductance is L (named after the French physicist Heinrich Lenz), while the unit is given in henries (H).

The simplest and most common example used to explain inductance is the coil. Although the phenomenon of inductance already occurs in a simple conductor, this effect is deliberately amplified by the construction of a coil. Figure 5.1 shows a cylindrical air coil with its magnetic flux density $\vec{B}$.

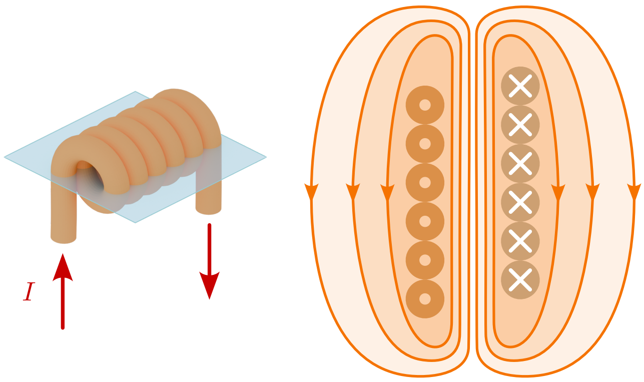


Figure 5.1: **Representation of the magnetic field of a cylindrical air coil.** It illustrates the field lines both inside and outside the coil and shows how the magnetic field concentrates along the axis of the coil and decreases outside the coil.

The topic of magnetic quantities is covered in detail in Module 6. To understand how a coil works and the resulting inductance, a brief introduction to the origin of magnetism will help.

As soon as current flows through a straight conductor, a magnetic field is created concentrically around the conductor. The magnetic field also changes depending on the current strength and direction. The right-hand rule serves as a mnemonic device for determining the direction and orientation of the magnetic field.

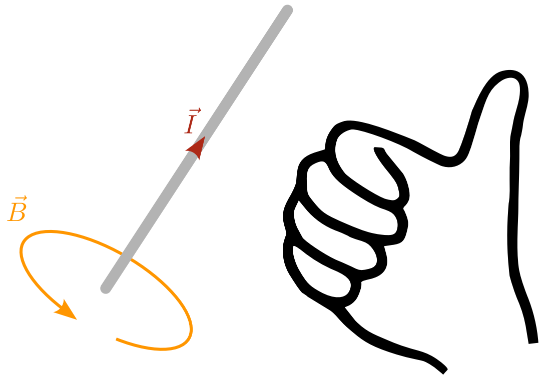


Figure 5.2: **Right-hand rule for determining the direction of the magnetic field.** A current-carrying conductor generates a magnetic field that is determined using the right-hand rule. If the thumb of the right hand points in the direction of the current, then the remaining fingers curve in the direction of the magnetic field lines.

According to the diagram 5.2, the magnetic field runs concentrically around the conductor in the direction of the fingertips, provided that the thumb is aligned in the direction of the technical current flow, i.e. opposite to the direction of electron flow. The magnetic field rotates from south to north and has a flux density $\vec{B}$ and a field strength $\vec{H}$. It should be noted that the magnetic flux density $\vec{B}$ behaves analogously to the electric field strength $\vec{E}$, and the magnetic field strength $\vec{H}$ behaves analogously to the electric flux density $\vec{D}$. Module 6 deals with the topic of magnetism in more detail. This module is primarily about getting to know the coil and inductance.

In the mid-19th century, British physicist and chemist Michael Faraday conducted experiments in which he moved a coil near a magnet or changed the current in an adjacent coil. He found that the magnitude of the induced voltage was proportional to the rate of change of the magnetic field.

$$U_\mathrm{i} \sim \frac{\mathrm{d}\phi}{\mathrm{d}t} \tag{5.1}$$

Faraday discovered that the proportionality factor depends on the number of windings in the coil.

$$U_\mathrm{i} = -N \cdot \frac{\mathrm{d}\phi}{\mathrm{d}t} \tag{5.2}$$

At around the same time, American physicist Joseph Henry discovered the phenomenon of self-induction, whereby an electric current in a coil can induce a voltage in the coil itself when the current is changed. Here, too, proportional behaviour was observed.

$$U_\mathrm{i} \sim \frac{\mathrm{d}I}{\mathrm{d}t} \tag{5.3}$$

This property of the coil, which arises from the storage of magnetic energy, is the inductance L, whose unit was named Henry [H] in honour of the physicist.

$$U_\mathrm{i} = -L \cdot \frac{\mathrm{d}I}{\mathrm{d}t} \tag{5.4}$$

The minus sign comes from the fact that, according to Lenz's law, the induced current counteracts its cause.

Analogous to the capacitor, the time-dependent behaviour for the coil can also be expressed as follows:

Voltage across the inductance

$$u_L(t) = L \cdot \frac{di_L(t)}{dt} \tag{5.5}$$

Current through the inductance

$$i_L(t) = \frac{1}{L} \cdot \int u_L(t)\, dt \tag{5.6}$$

5.2 The coil as a component

With knowledge of the magnetic field, the structure and function of the coil can now be explained. A simple example of this is the cylindrical air coil. When a conductor is formed into a conductor loop, the magnetic field is formed in such a way that it passes through the loop opening. If several conductor loops are wound one behind the other, the magnetic field is amplified by the number of turns. This arrangement of wound conductor loops results in the cylindrical air coil shown in Figure 5.3.

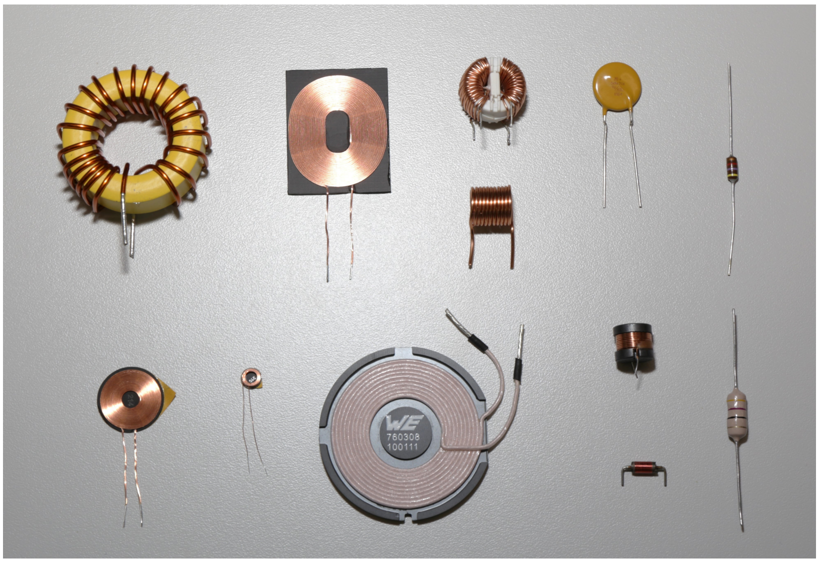


Figure 5.3: **Various types and designs of coils.** These include cylindrical coils, toroidal coils, iron core coils and air coils. A larger number of turns increases the concentration of the magnetic field and increases the inductance of the coil. An iron core can also be used to bundle the magnetic field lines and improve efficiency.

This coil owes its name to its cylindrical shape in the longitudinal direction and the medium of air inside its windings. When a current flows through the coil, the magnetic fields of the individual windings superimpose to form a stronger magnetic field. The strength depends on the number of turns and the current, as well as the distance to the conductor. Depending on the orientation of the current and the windings, a cylinder air coil produces a north pole on one side and a south pole on the other. The course of the magnetic field lines in a cylindrical air coil corresponds to the illustration 5.1.

What all coils have in common is their ability to utilise the property of inductance, as shown in the following circuit diagram 5.5. However, this circuit diagram only represents the idealised, usable inductance.

Figure 5.4: **Circuit element for coil.** European (left) and American (right).

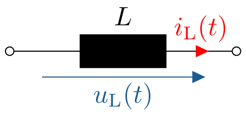


Figure 5.5: **Circuit symbol for an ideal coil.**

For a realistic simulation of the circuit, it is important to describe the coil as a component in its entirety. This is done with the help of an equivalent circuit diagram, which also simulates the unwanted, i.e. parasitic, properties of the component. Figure 5.6 shows such an equivalent circuit diagram.

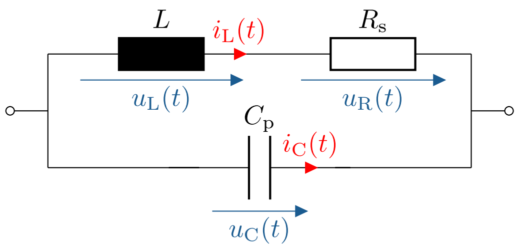


Abbildung 5.6: **Equivalent circuit diagram of a real coil.** The ideal coil L is supplemented by a series resistor R_s and a parallel capacitor C_p to account for parasitic effects.

The equivalent circuit diagram of a coil takes into account not only the usable inductance, but also the parasitic properties that occur in real coils. These parasitic properties arise, for example, from the capacitances between the windings and the ohmic losses in the material. The modelled equivalent circuit diagram makes it possible to take these effects into account in the circuit design and to better predict the switching behaviour.

Key point:

The coil is a desperate attempt to replicate inductance.

5.3 Permeability

Similar to the case of the capacitor and capacitance, the strength of the magnetic field also varies depending on the material. What are the dielectric and permittivity in the case of the capacitor are referred to as ferromagnetic material and permeability in the case of the coil. An example of a coil with a ferromagnetic material is the toroidal coil 5.7..

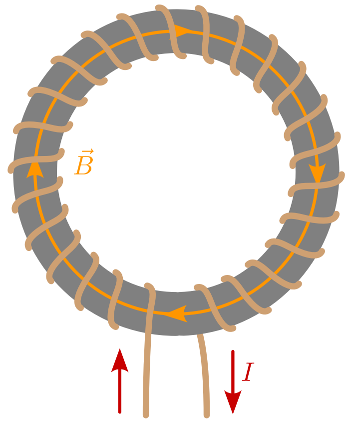


Figure 5.7: **Toroidal coil with iron core.** The iron core strengthens the magnetic field and increases the inductance of the coil. This design minimises stray fields and improves efficiency by concentrating the magnetic field within the core.

The toroidal coil is a ferrite ring wrapped with wire. Alignment takes place at the atomic level within the material, which leads to a strengthening of the magnetic field. The strength of the magnetic flux density $\vec{B}$ is determined not only by the dimensions and current strength, but also by the material properties, in particular the permeability.

The following table 5.1 shows the permeability coefficient of some common materials:

Material	$\boldsymbol{\mu_r}$
Type I superconductor	0
Vacuum	£1
Air (at STP)	£1.00000037
Copper	£0.999994
Gold	£0.999964
Aluminium	£1.000022
Iron	$300 - 10000$
Ferrite	$4 - 15000$

Table 5.1: **Relative permeability coefficient of various materials**: This dimensionless number indicates how strongly the magnetic field in a material is affected in comparison to a vacuum. It is a measure of the material's ability to store or conduct magnetic flux. Materials with a higher permeability number increase the inductance of coils.

In order to fully describe the properties of ferromagnetic material, in addition to the permeability coefficient, the magnetic field constant μ_0 is also required. This refers to the behaviour of the magnetic field in a vacuum. While the permeability coefficient μ_r is dimensionless, the magnetic field constant has the unit H/m. It is multiplied by the relative permeability of the material, which together gives the permeability constant μ..

$$[\mu] = 1\,\frac{\text{Henry}}{\text{m}} = 1\,\frac{\text{H}}{\text{m}} = 1\,\frac{\text{Vs}}{\text{Am}}$$

$$\mu = \mu_\text{r} \cdot \mu_0 \tag{5.7}$$

$$\begin{aligned} \mu &: \text{Permeability constant} \\ \mu_0 &: \text{Magnetic field constant } (\approx 4\pi \cdot 10^{-7}) \\ \mu_\mathrm{r} &: \text{Relative permeability coefficient} \end{aligned}$$

With the help of the permeability, the current, the number of turns and the length of the coil, the magnetic flux density $\vec{B}$ can be calculated using the unit Tesla T.

$$\vec{B} = \mu \cdot \vec{H} \tag{5.8}$$

$$[B] = \frac{\mathrm{H/m \cdot A \cdot 1}}{\mathrm{m}} = \frac{\mathrm{H \cdot A}}{\mathrm{m^2}} = \frac{(\mathrm{V \cdot s} \,/\, \cancel{\mathrm{A}}) \cdot \cancel{\mathrm{A}}}{\mathrm{m^2}} = \frac{\mathrm{V \cdot s}}{\mathrm{m^2}} = \mathrm{T\ (Tesla)}$$

$$B = \frac{\mu \cdot I \cdot N}{l} \tag{5.9}$$

$$\begin{aligned} \mu &: \text{Permeability constant} \\ I &: \text{Electrical currente} \\ N &: \text{Number of turns} \\ l &: \text{Length of the magnetic circuit} \end{aligned}$$

The inductance itself also depends on the design and the materials used.

$$[L] = 1\,\mathrm{Henry} = 1\,\mathrm{H} = 1\,\frac{\mathrm{Vs}}{\mathrm{A}}$$

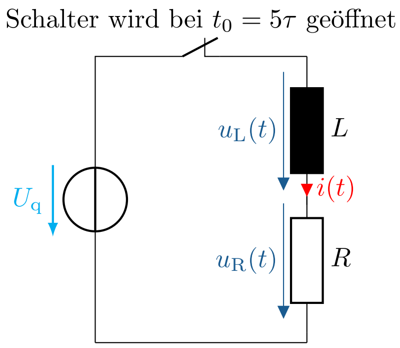
$$L = \frac{\mu \cdot N^2 \cdot A}{l} \tag{5.10}$$

5.4 The magnetic field strength

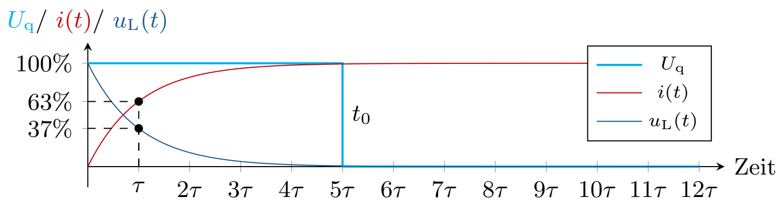
$$\vec{B} = \mu \cdot \vec{H} \tag{5.11}$$

5.5 Switching behaviour of a coil

To better understand how a coil works, let us now consider a time-varying signal, in this case the switch-on and switch-off process. The basic principle of inductance is based on the fact that a time-varying current generates a time-varying magnetic field. This time-varying magnetic field in turn induces an induction current in the coil itself. However, due to Lenz's law, the induction current acts against its cause, which prevents a sudden increase in current.

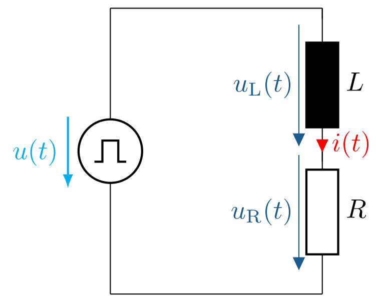


(a) Representation of the series connection consisting of an inductance L, a resistor R and a switch.

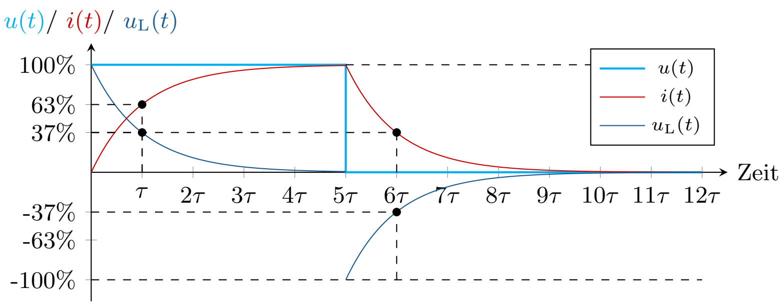


(b) Voltage and current waveform across the inductance for a DC voltage that is switched on at time $t = 0$ and switched off at time t_0.

Figure 5.8: **The voltage and current curve across the inductance.** The diagram shows the reaction of the inductance to the sudden change in voltage and illustrates the typical charging behaviour.



(a) Representation of the series connection consisting of an inductance L and a resistor R connected to a rectangular voltage source.



(b) Voltage and current waveform across the inductance with a square wave voltage.

Figure 5.9: **Voltage and current curve across the inductance.** The diagram shows the reaction of the inductance to a sudden change in voltage and illustrates the typical charging and discharging behaviour of the inductance.

In practical applications, inductance enables the construction of components such as transformers, chokes and inductors, which are found in a wide range of electrical and electronic devices. From signal filtering and energy transmission to the control of electromagnetic interference, inductance is a key element in modern electrical engineering.

Example 5.1: Calculation of inductance

- Calculation of inductance L
- Calculation of magnetic flux density $\vec{B}$
- Energy stored in the magnetic field E_{m} ?

A Exercise tasks

A.1 Material dependence of resistance R

A copper wire has the following properties:

- Length: $l = 5\,mathrmm$
- Cross-sectional area: $A = 1\,\mathrm{mm}^2$
- Specific resistance of copper: $\rho = 1.68 \times 10^{-8}\,\Omega\,\mathrm{m}$

Calculate R of the wire.

A.2 Electrical resistance

A spherical hollow structure consists of two ideally conductive shells – an inner one with radius a and an outer one with radius b.

The space between them is completely filled with a homogeneous, conductive material that has electrical conductivity κ (in $\mathrm{S\,m^{-1}}$).

A voltage

$$U = \varphi_1 - \varphi_2$$

is applied between the conductive shells by a voltage source.

The current can spread unhindered due to the complete filling and ideal contacts.

Calculate the electrical resistance R between the two spherical shells as a function of a, b and κ.

A.3 Electrical resistance of a coated wire

A round aluminium wire has a radius of $a = 0.5\,\mathrm{mm}$. It is coated with a thin layer of gold. $b = 0.1\,\mathrm{mm}$ Thickness coated.

The following values apply to the specific electrical conductivity and temperature coefficient of the two metals:

- **Aluminium:**
$$\kappa_{\mathrm{Alu}} = 35\,\mathrm{m}/(\Omega\mathrm{mm}^2), \quad \alpha_{\mathrm{Alu}} = 3{,}5 \cdot 10^{-3}\,\mathrm{K}^{-1}$$
- **Gold:**
$$\kappa_{\mathrm{Gold}} = 44\,\mathrm{m}/(\Omega\mathrm{mm}^2), \quad \alpha_{\mathrm{Gold}} = 3{,}6 \cdot 10^{-3}\,\mathrm{K}^{-1}$$

Given: Wire Length: $l = 1.5\,\mathrm{m}$ Temperature: $T = 25\,°\mathrm{C}$ (Initial temperature)

a) Calculate the electrical DC resistance of the aluminium core and gold coating at $T = 25\,°\mathrm{C}$.

b) How does the respective direct current resistance change when the temperature rises to $T =$ 90°C?

A.4 Real power sources

A real current source has an open-circuit voltage $U_0 = 9\,\mathrm{V}$ and an internal resistance $R_\mathrm{i} = 1\,\Omega$.

a) How large is the terminal voltage when a load resistance of $R_l = 4\,\Omega$ is connected? b) How large is the current through the load?

A.5 Voltage source

Given is a network consisting of a voltage source with $U = 10\,\mathrm{V}$, a series resistor $R_1 = 2\,\Omega$ and a parallel resistor $R_2 = 4\,\Omega$.

Calculate currents and voltages in the network.

A.6 Capacitance and capacitor

Part 1: Single capacitor

Given:

$$C = 10\,\mathrm{pF}, \quad U = 5\,\mathrm{V}$$

a) Calculate the load Q

b) Calculate the energy stored in the capacitor.

Part 2: Two capacitors in series

Two capacitors C_1 and C_2 are connected in series.

$$C_1 = 4\,\mathrm{\mu F}, \quad C_2 = 6\,\mathrm{\mu F}, \quad U_\mathrm{ges} = 12\,\mathrm{V}$$

c) Calculate the total capacity C_ges

d) How great is the excitement about C_1?

A.7 Inductance and coil

Part 1: Energy in the coil

Given is a coil with the inductance

$$L = 2\,\mathrm{H}$$

and a current

$$I = 3\,\mathrm{A}.$$

a) Calculate the energy stored in the magnetic field of the coil.

b) How does the energy change when the current is reduced to $I = 0.5\,\mathrm{A}$?

Part 2: Inductance of a cylindrical coil

A cylindrical coil has

$$N = 500 \text{ Wicklungen}, \quad l = 20\,\mathrm{cm}, \quad A = 5\,\mathrm{cm}^2$$

The inside of the coil is filled with air.

Calculate the inductance L of the coil.

A.8 Inductance, magnetic flux density and magnetic field energy

An air-filled, long cylindrical coil conductor has the following properties:

- Number of turns: $N = 500$
- Length of the coil: $l = 0.5\,\mathrm{m}$
- Cross-sectional area: $A = 4 \cdot 10^{-4}\,\mathrm{m}^2$
- Current: $I = 2.5\,\mathrm{A}$

Given: Magnetic field constant: $\mu_0 = 4\pi \cdot 10^{-7}\,\mathrm{H\,m^{-1}}$

a) Calculate the magnetic flux density $\vec{B}$ inside the coil.

b) Determine the inductance L of the coil.

c) Calculate the energy E_m stored in the magnetic field.

B Solutions to the exercises

B.1 Material dependence of resistance R

$$R = \rho \cdot \frac{l}{A} = 1{,}68 \cdot 10^{-8} \cdot \frac{5}{1 \cdot 10^{-6}} = 0.084\,\Omega$$

B.2 Electrical resistance

Since the system is radially symmetric, the current flows radially from the inside to the outside (or vice versa), and the electric field depends only on the radius r.

The current density is given by Ohm's law in differential form:

$$\vec{j}(r) = \kappa \cdot \vec{E}(r)$$

The area through which the current flows at radius r is the surface area of a sphere:

$$A(r) = 4\pi r^2$$

The total current is constant across all spherical surfaces:

$$I = j(r) \cdot A(r) = \kappa \cdot E(r) \cdot 4\pi r^2 \quad \Rightarrow \quad E(r) = \frac{I}{4\pi\kappa r^2}$$

Calculate voltage:

The voltage is calculated using the integral over the electric field:

$$U = \int_a^b E(r)\,dr = \int_a^b \frac{I}{4\pi\kappa r^2}\,dr = \frac{I}{4\pi\kappa}\int_a^b \frac{1}{r^2}\,dr = \frac{I}{4\pi\kappa}\left[-\frac{1}{r}\right]_a^b$$

$$U = \frac{I}{4\pi\kappa}\left(\frac{1}{a} - \frac{1}{b}\right)$$

Calculate resistance:

Since $U = R \cdot I$ applies, it follows that

$$R = \frac{U}{I} = \frac{1}{4\pi\kappa}\left(\frac{1}{a} - \frac{1}{b}\right)$$

The electrical resistance between the two shells is therefore:

$$\boxed{R = \frac{1}{4\pi\kappa}\left(\frac{1}{a} - \frac{1}{b}\right)}$$

B.3 Electrical resistance of a coated wire

Radius aluminium core: $a = 0.5\,\mathrm{mm}$
Thickness of the gold layer: $b = 0.1\,\mathrm{mm}$
$\Rightarrow$ Total radius: $r = a + b = 0.6\,\mathrm{mm}$
Length of the cable: $l = 1.5\,\mathrm{m} = 1500\,\mathrm{mm}$
Aluminium: $\kappa_{\mathrm{Alu}} = 35\,\mathrm{m/(\Omega mm^2)}$, $\alpha_{\mathrm{Alu}} = 3{,}5 \times 10^{-3}\,\mathrm{K^{-1}}$
Gold: $\kappa_{\mathrm{Gold}} = 44\,\mathrm{m/(\Omega mm^2)}$, $\alpha_{\mathrm{Gold}} = 3{,}6 \times 10^{-3}\,\mathrm{K^{-1}}$
Change in temperature:
$T_1 = 25\,^\circ\mathrm{C} = 298\,\mathrm{K}$
$T_2 = 90\,^\circ\mathrm{C} = 363\,\mathrm{K}$
$\Delta T = T_2 - T_1 = 65\,\mathrm{K}$

Calculate cross-sectional areas

Aluminium core: $A_{\mathrm{Alu}} = \pi a^2 = \pi \times (0.5)^2 = \pi \times 0.25 = 0.7854\,\mathrm{mm^2}$
Gold mantle (ring area between $r = 0.6$ mm and $a = 0.5$ mm):
$A_{\mathrm{Gold}} = \pi(r^2 - a^2) = \pi(0.6^2 - 0.5^2) = \pi(0.36 - 0.25) = \pi \times 0.11 \approx 0.3456\,\mathrm{mm^2}$

Resistance at $25\,^\circ\mathrm{C}$

Formula:

$$R = \frac{l}{\kappa \cdot A}$$

$$R_{\mathrm{Alu}} = \frac{1500}{35 \times 0.7854} = \frac{1500}{27.489} \approx 54.58\,\mathrm{m\Omega}$$
$$R_{\mathrm{Gold}} = \frac{1500}{44 \times 0.3456} = \frac{1500}{15.206} \approx 98.61\,\mathrm{m\Omega}$$

Resistance at $90\,^\circ\mathrm{C}$

Temperature-dependent resistance:

$$R(T) = R_0 \cdot (1 + \alpha \cdot \Delta T)$$

$$R_{\text{Alu},90^\circ C} = 54.58 \times (1 + 3.5 \times 10^{-3} \times 65) = 54.58 \times 1.2275 \approx 66.97\,\text{m}\Omega$$
$$R_{\text{Gold},90^\circ C} = 98.61 \times (1 + 3.6 \times 10^{-3} \times 65) = 98.61 \times 1.234 \approx 121.64\,\text{m}\Omega$$

Material	resistance at 25 °C	resistance at 90 °C
Aluminium	54.58 mΩ	66.97 mΩ
Gold	98.61 mΩ	121.64 mΩ

B.4 Real current sources

a) Terminal voltage $U_\mathbf{k}$:

The terminal voltage is calculated using the voltage divider:

$$U_\text{k} = U_0 \cdot \frac{R_\text{l}}{R_\text{i} + R_\text{l}} = 9\,\text{V} \cdot \frac{4\,\Omega}{1\,\Omega + 4\,\Omega} = 9 \cdot \frac{4}{5} = 7.2\,\text{V}$$

b) Current through the load I:

$$I = \frac{U_0}{R_\text{i} + R_\text{l}} = \frac{9}{1+4} = \frac{9}{5} = 1.8\,\text{A}$$

B.5 Voltage source

$$R_\text{ges} = R_1 + R_2 = 2 + 4 = 6\,\Omega, \quad I = \frac{U}{R_\text{ges}} = \frac{10}{6} = 1.67\,\text{A}$$

$$U_{R_1} = I \cdot R_1 = 1.67 \cdot 2 = 3.33\,\text{V}, \quad U_{R_2} = U - U_{R_1} = 6.67\,\text{V}$$

$$I_{R_2} = \frac{U_{R_2}}{R_2} = \frac{6.67}{4} = 1.67\,\text{A}$$

B.6 Capacitance and capacitor

a) Load:

$$Q = C \cdot U = 10 \cdot 10^{-12} \cdot 5 = 50\,\text{pC}$$

b) Stored energy:

$$E = \frac{1}{2}CU^2 = \frac{1}{2} \cdot 10 \cdot 10^{-12} \cdot 25 = 125\,\text{pJ}$$

c) Total capacity of the series connection:

$$\frac{1}{C_\text{ges}} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{4} + \frac{1}{6} = \frac{5}{12} \Rightarrow C_\text{ges} = 2.4\,\mu\text{F}$$

d) tension over C_1:

$$Q = C_{\text{ges}} \cdot U_{\text{ges}} = 2.4 \cdot 10^{-6} \cdot 12 = 28.8\,\mu\text{C}$$
$$U_1 = \frac{Q}{C_1} = \frac{28.8 \cdot 10^{-6}}{4 \cdot 10^{-6}} = 7.2\,\text{V}$$

B.7 Inductance and coil

a) Energy at $I = 3\,\text{A}$:

$$E = \frac{1}{2} L I^2 = \frac{1}{2} \cdot 2 \cdot 3^2 = 9\,\text{J}$$

b) Energy at $I = 0.5\,\text{A}$:

$$E = \frac{1}{2} \cdot 2 \cdot 0.5^2 = 0.25\,\text{J}$$

c) Calculation of inductance L of the cylindrical coil:

First, convert the units:

$$l = 20\,\text{cm} = 0.2\,\text{m}, \quad A = 5\,\text{cm}^2 = 5 \times 10^{-4}\,\text{m}^2$$

The inductance of an air-filled cylindrical coil is calculated by:

$$L = \mu_0 \frac{N^2 A}{l}$$

with the magnetic field constant

$$\mu_0 = 4\pi \times 10^{-7}\,\text{H}\,\text{m}^{-1}$$

Insert:

$$L = 4\pi \times 10^{-7} \cdot \frac{500^2 \cdot 5 \times 10^{-4}}{0.2} = 7.85 \times 10^{-2}\,\text{H} = 78.5\,\text{mH}$$

B.8 Inductance, magnetic flux density and magnetic field energy

a) Magnetic flux density $\vec{B}$:

The following applies to a long, air-filled cylinder coil:

$$B = \mu_0 \cdot \frac{N}{l} \cdot I$$

Applying the values:

$$B = 4\pi \cdot 10^{-7} \cdot \frac{500}{0.5} \cdot 2.5 = 4\pi \cdot 10^{-7} \cdot 1000 \cdot 2.5$$
$$B = 4\pi \cdot 10^{-7} \cdot 2500 = \pi \cdot 10^{-3}\,\text{T}$$
$$\Rightarrow B \approx 3.14 \cdot 10^{-3}\,\text{T} = 3.14\,\text{mT}$$

b) Inductance L:

The inductance of a long air-filled coil is calculated as follows:

$$L = \mu_0 \cdot \frac{N^2 \cdot A}{l}$$

Applying the values:

$$L = 4\pi \cdot 10^{-7} \cdot \frac{500^2 \cdot 4 \cdot 10^{-4}}{0.5}$$

$$L = 4\pi \cdot 10^{-7} \cdot \frac{250000 \cdot 4 \cdot 10^{-4}}{0.5} = 4\pi \cdot 10^{-7} \cdot \frac{100}{0.5} = 4\pi \cdot 10^{-7} \cdot 200$$

$$L = 8\pi \cdot 10^{-5}\,\mathrm{H} \Rightarrow L \approx 2.51 \times 10^{-4}\,\mathrm{H} = 251\,\mu\mathrm{H}$$

c) Energy stored in the magnetic field E_m:

The energy in the magnetic field of a coil is calculated as follows:

$$E_m = \frac{1}{2} L I^2$$

Insert:

$$E_m = \frac{1}{2} \cdot 2.51 \cdot 10^{-4} \cdot (2.5)^2 = 0.5 \cdot 2.51 \cdot 10^{-4} \cdot 6.25$$

$$E_m = 7.84 \cdot 10^{-4}\,\mathrm{J} = 0.784\,\mathrm{mJ}$$